

CM4AI AnVIL Cell Map Prediction

DREAM Challenge

A single human cell contains thousands of protein types, each precisely positioned to keep the cell alive and working. Where a protein is placed and how it moves in the cell shapes everything from cell structure to how cells sense and respond to their environment. Yet today's imaging tools can visualize only tens of proteins in the same cell. We need technologies that integrate such data to learn and simulate protein localization. This will enable us to build complete protein maps of cells and model how cellular protein architecture shifts in health and disease.

Help pioneer *in silico* microscopy and build "virtual cells": join this challenge to develop generative AI models that predict where a protein localizes, using reference microscopy images and the protein's amino acid sequence as input.

WHEN

Starts

October 19, 2026

Ends

February 1, 2027

PRIZES

Cash

GCP credits

1st

\$25K

\$40K

2nd

\$15K

\$20K

3rd

\$10K

\$10K

JOIN THE CHALLENGE

WEBSITE

synapse.org/CM4AI_DREAM

